Relation Algebra

§1. Notation

Traditionaly Adjunction F -| G is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$F(x) \leq F(x) \implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x)))$$
$$G(y) \leq G(y) \implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y)$$

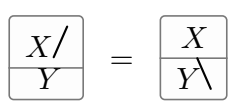
By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

(1a)

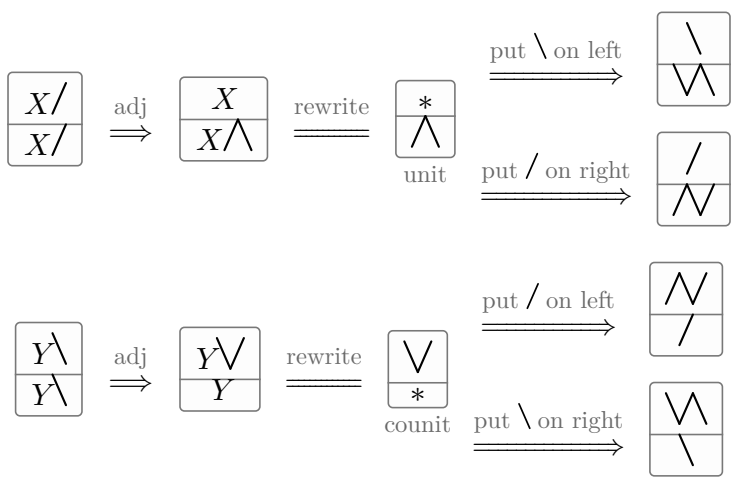
1 • write F as $/$, G as $\backslash$

2 • render $a \leq b$ as $\frac{a}{b}$

3 • **rewrite** $X \leq XF$ as $1 \leq F$



adj



now we have $\diagup\!\!\diagdown = /$ and $\diagdown\!\!\diagup = \backslash$, and you just discovered string diagrams.

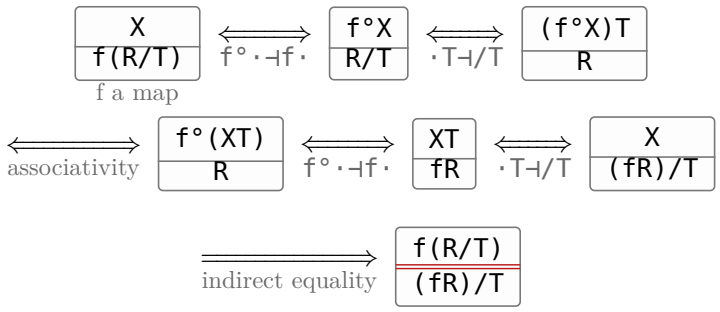
§1.1. Adjunctions

(1.1a)

	F⊣G	F's type	monad FG	1⊑FG	GF⊑1	F preserves ∪	G preserves ∩	FGF=F	GFG=G
1	◁⊣▷	A→ A⊗A	◁▷=𝟙	𝟙⊑◁▷	▷◁⊑𝟙	(R ∪ S)◁= R◁ ∪ S◁	(R∩S)▷= R▷∩S▷	◁▷◁=◁	▷◁▷=▷
2	⊸⊣⊸	A→𝕀	⊸⊸=⊤	𝟙⊑⊸⊸	⊸⊸⊑𝟙	(R ∪ S)⊸= R⊸ ∪ S⊸	(R∩S)⊸= R⊸∩S⊸	⊸⊸⊸=⊸	⊸⊸⊸=⊸
3	°⊣°	(A→B)→ (B→A)	R°° = R	R°° = R	R°° = R	(R ∪ S)° = S° ∪ R°	(R ∩ S)° = R° ∩ S°	R°° = R	R°° = R
4	⊸◁⊣▷⊸	(X⊗A→Y)→ (X→Y⊗A)	𝟙	(⊸◁⊗𝟙) (𝟙⊗▷⊸)=𝟙	(𝟙⊗⊸◁) (▷⊸⊗𝟙)=𝟙	=	=	=	=
5	▷⊸⊣⊸◁	(X→Y⊗A)→ (X⊗A→Y)	𝟙	(𝟙⊗⊸◁) (▷⊸⊗𝟙)=𝟙	(⊸◁⊗𝟙) (𝟙⊗▷⊸)=𝟙	=	=	=	=
6	Δ⊣∩	(A→B)→ (A→B)²	R ∩ R = R	R ∩ R = R	R ∩ S ⊑ R	Δ(R ∪ S)= ΔR ∪ ΔS	(R∩T)∩(S∩U)= (R∩S)∩(T∩U)	R ∩ R = R	R ∩ R = R
7	∪⊣Δ	(A→B)²→ (A→B)	(R,S)↦ (R ∪ S,R ∪ S)	R ⊑ R ∪ S	R ∪ R = R	(R ∪ T) ∪ (S ∪ U)= (R ∪ S) ∪ (T ∪ U)	Δ(R∩S)= ΔR∩ΔS	R ∪ R = R	R ∪ R = R
8	𝟘⊣!	{*}→ (A→B)			𝟘 ⊑ R				
9	·S⊣/S	(A→B)→ (A→C)	S/S	R⊑(RS)/S	(T/S)S⊑T	(S ∪ T) R = (S R) ∪ (T R)	(R ∩ R') / S = (R / S) ∩ (R' / S)	((RS)/S)S =RS	((T/S)S)/S =T/S
10	S·⊣S\	(B→C)→ (A→C)	S\S	R⊑S\(SR)	S(S\T)⊑T	R (S ∪ T) = (R S) ∪ (R T)	S \ (R ∩ R') = (S \ R) ∩ (S \ R')	S(S\(SR)) =SR	S\(S(S\T)) =S\T
11	R∩⊣R⇒	(A→B)→ (A→B)	X↦R⇒(X∩R)	X⊑R⇒(X∩R)	R∩(R⇒Y)⊑Y	R∩(X ∪ Y)= (R∩X) ∪ (R∩Y)	R ⇨ S ∩ T = (R ⇨ S) ∩ (R ⇨ T)	R∩(R⇒(X∩R)) =X∩R	R⇒(R∩(R⇒Y)) =R⇒Y
12	𝒟⊣·⊤	(A→B)→ Cor A	R↦(𝒟R)⊤	R⊑(𝒟R)⊤	𝒟(A⊤)⊑A	𝒟(R ∪ S)= 𝒟R ∪ 𝒟S	(A∩B)⊤= A⊤∩B⊤	𝒟((𝒟R)⊤) =𝒟R	(𝒟(A⊤))⊤ =A⊤
13	ℛ⊣⊤·	(A→B)→ Cor B	R↦⊤(ℛR)	R⊑⊤(ℛR)	ℛ(⊤A)⊑A	ℛ(R ∪ S)= ℛR ∪ ℛS	⊤(A∩B)= ⊤A∩⊤B	ℛ(⊤(ℛR)) =ℛR	⊤(ℛ(⊤A)) =⊤A
14	·f⊣·f°	(A→B)→ (A→C)	ff°	Map(f) ⇒ 𝟙 ⊑ f f°	f°f⊑𝟙	(S ∪ T) R = (S R) ∪ (T R)	Simple(g) ⇒ (X ∩ Y) g° = (X g°) ∩ (Y g°)	ff°f=f	f°ff°=f°
15	f°·⊣f·	(A→C)→ (B→C)	ff°	Map(f) ⇒ 𝟙 ⊑ f f°	f°f⊑𝟙	R (S ∪ T) = (R S) ∪ (R T)	f(X∩Y)= fX∩fY	ff°f=f	f°ff°=f°
16	i⊣E	Map↪Rel	E	$\frac{𝟙}{∋}$:A→PA	∋:PB→B	—	—	$\frac{∋}{∋}$=𝟙	$\frac{R}{∋}$∋=R
17	·∋⊣ $\frac{}{∋}$	Map(A,PB)→ (A→B)	𝟙	$\frac{f∋}{∋}$=f	$\frac{R}{∋}$∋=R	—	—	$\frac{f∋}{∋}$∋ =f∋	$\frac{\frac{R}{∋}∋}{∋}$ =$\frac{R}{∋}$
18	$\frac{}{∋}$ ⊣·∋	(A→B)→ Map(A,PB)	𝟙	$\frac{R}{∋}$∋=R	$\frac{f∋}{∋}$=f	—	—	$\frac{\frac{R}{∋}∋}{∋}$ =$\frac{R}{∋}$	$\frac{f∋}{∋}$∋ =f∋

§1.2. Composing adjunctions

(1.2a)




```
Map(f) ⇒ f (R / S) = (f R) / S
```

(1.2b)

		·T	T·	·g	g°·	T∩	°	⊶◁	Δ
1	**·S**	X / (S₁ S₂) = X / S₂ / S₁	S \ (X / W) = (S \ X) / W	R/(Sg)= (Rg°)/S	Map(f) ⇒ f (R / S) = (f R) / S	—	(Y/S)°= S°\(Y°)	(RS)^= R^(S⊗𝟙)	(R ∩ R') / S = (R / S) ∩ (R' / S)
2	**S·**	S \ (X / W) = (S \ X) / W	(S₁ S₂) \ X = S₂ \ (S₁ \ X)	S\(Rg°)= (S\R)g°	(g°S)\R= S\(gR)	—	(S\Y)°= Y°/S°	((S⊗𝟙)R)^= SR^	S \ (R ∩ R') = (S \ R) ∩ (S \ R')
3	**·f**	Map(f) ⇒ R / (f S) = (R / S) f°	T\(Rf°)= (T\R)f°	(R S)° = S° R°			(R S)° = S° R°		Simple(g) ⇒ (X ∩ Y) g° = (X g°) ∩ (Y g°)
4	**f°·**	Map(f) ⇒ f (R / S) = (f R) / S	(Tf°)\R= f(T\R)	—	(R S)° = S° R°		(R S)° = S° R°		f(T₁∩T₂)= fT₁∩fT₂
5	**R∩**	—	—	—	—	R ⇨ S ⇨ T = R ∩ S ⇨ T	(R⇒Y)°= R°⇒(Y°)	—	R ⇨ S ∩ T = (R ⇨ S) ∩ (R ⇨ T)
6	**°**	(Y/T)°= T°\(Y°)	(T\Y)°= Y°/T°	(R S)° = S° R°	(R S)° = S° R°	(T⇒Y)°= T°⇒(Y°)	R°° = R	(R^)°= (R°⊗𝟙) (𝟙⊗▷⊸) both bends: the ° section's display	(T₁∩T₂)°= T₁°∩T₂°
7	**⊶◁**	(RT)^= R^(T⊗𝟙)	((T⊗𝟙)R)^= TR^	—	—	—	(R^)°= (R°⊗𝟙) (𝟙⊗▷⊸) both bends: the ° section's display	—	—
8	**∪**	(S ∪ T) R = (S R) ∪ (T R)	R (S ∪ T) = (R S) ∪ (R T)	(S ∪ T) R = (S R) ∪ (T R)	R (S ∪ T) = (R S) ∪ (R T)	R ∩ (S ∪ T) = R ∩ S ∪ R ∩ T	(R ∪ S)° = S° ∪ R°		
9	**0**	0 R = 0	R 0 = 0	0 R = 0	R 0 = 0	R ∩ 0 = 0	0° = 0		

§1.3. Adjoint triples

Beyond `∪⊣Δ⊣∩`, which §1.1 already carries as two rows, the table holds one adjoint triple with content: `∃_f⊣f*⊣∀_f` along a map `f`, once on each side of the composite. `f : A→B` throughout this subsection.

(1.3a)

$$\cdot f \dashv \cdot f^\circ \dashv /f^\circ$$
$$f^\circ \cdot \dashv f \cdot \dashv f\backslash$$

The first two links of each chain are the map shunting rules, and the third is §1.1′s division row read at a chosen divisor: `·S⊣/S` at `S:=f°`, and `S·⊣S\` at `S:=f`.

The third link is not the first over again. For a map `f`, `/f` collapses to `·f°`; being a map is exactly what that collapse spends `Map(f) ⇒ R / (f S) = (R / S) f°` — `f°` is not a map, so `/f°` does not collapse in turn, and the three operators stay distinct. On `Rel(𝟙,A)=PA` they are the image triple:

(1.3b)

	operator	acts by	name
1	`·f`	`S↦f[S]`	direct image, `∃_f`
2	`·f°`	`T↦f⁻¹[T]`	inverse image, `f*`
3	`/f°`	`S↦{b : f⁻¹(b)⊆S}`	`∀_f`

§1.1′s `𝒟⊣·⊤` is this same chain along the projection `A⊗B→A`, read through `Rel(A,B)=P(A⊗B)`: `𝒟R={(a,a) : ∃b. aRb}`, `A⊤` is the relation that ignores `b` altogether, and the third link is `R↦𝟙∩R/⊤={(a,a) : ∀b. aRb}`.

(1.3c)

$$\mathcal{D} \dashv \cdot \top \dashv \mathbb{1} \cap \cdot /\top$$

Three is where it stops, and one map breaks both ends. Take `A={a₁,a₂}` and `B={b}`: `f[{a₁}]∩f[{a₂}]={b}` while `f[{a₁}∩{a₂}]=∅`, so `·f` does not preserve meets and has no left adjoint, and the same two subsets give `∀_f({a₁} ∪ {a₂})={b}` against `∀_f{a₁} ∪ ∀_f{a₂}=∅`, so `/f°` does not preserve joins and has no right adjoint. A monic `f` restores the binary case and no more — `⊤f` still reaches only `im f`, and `∀_f𝟘=B\im f` is still not `𝟘`. The chain extends only when `f°` is a map as well, and then `/f°=·f°°=·f` and it repeats forever. For an `S` with neither `S` nor `S°` a map there is no triple at all: `·S⊣/S` is two links and stops.

The rows that chain forever say nothing by it: `°` is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and `°⊣°⊣°` has no end, and §1.1′s other row of that kind, `⊶◁⊣▷⊶`, goes the same way.

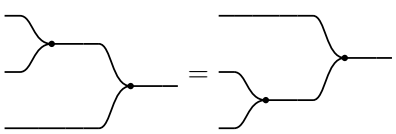
§2. Relations

(2a)

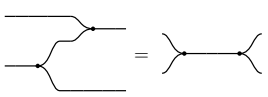
Rel is a poset-enriched category with $(\otimes, \circ)$ where $\otimes$ is commutative cartisian product, and converse $\circ \dashv \circ$, and

C: `(▷ : A⊗A→A,⊸ : 𝕀→A)` a commutative monoid with `C°⊣C`. We write `C°` as `(◁ : A→A⊗A,⊸ : A→𝕀)`

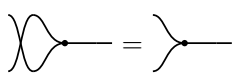
(2b)



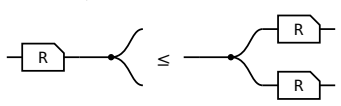
▷ associative



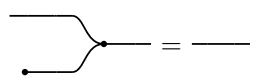
Frobenius, one half — the other is its $\circ$



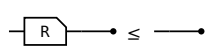
▷ commutative



R ◁ ≤ ◁ (R ⊗ R)



⊸ is its unit



R ⊸ ≤ ⊸

§2.1. ∀ object A, (A,◁,⊸)⊣(A,▷,⊶)

(2.1a)



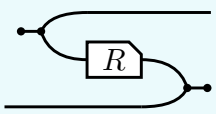
▷ ◁ ≤ 𝟙 | 𝟙 ≤ ◁ ▷ | ⊶ ⊸ ≤ 𝟙 | 𝟙 ≤ ⊸ ⊶

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In `Rel` it reads `{(a,a)}⊆A×A` — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§3. ° : $\mathcal{C}^{op} \longrightarrow \mathcal{C}$ is a 2 functor

(3a)

° is primitive, part of the data the first section lists. It turns both of R's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R°, not its definition:



$$R^\circ = (\multimap\!\triangleleft \otimes \mathbb{1})(\mathbb{1} \otimes R \otimes \mathbb{1})(\mathbb{1} \otimes \triangleright\!\multimap)$$

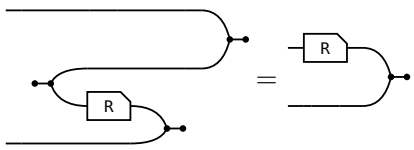
where $\multimap\!\triangleleft : \mathbb{I} \to a \otimes a$ opens a pair of wires out of nothing and $\triangleright\!\multimap : b \otimes b \to \mathbb{I}$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\multimap$ are $\triangleright^\circ$ and $\multimap^\circ$. The laws of ° are that it is a contravariant 2-functor $^\circ : \mathcal{C}^{\circ p} \to \mathcal{C}$:

(i) $\mathbb{1}^\circ = \mathbb{1}$ (ii) $(RS)^\circ = S^\circ R^\circ$ (iii) $(R \otimes S)^\circ = R^\circ \otimes S^\circ$ (iv) $R \leq S$ implies $R^\circ \leq S^\circ$

§3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:

(3.1a)

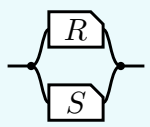


`R°` is DEFINED as the bending of `(R⊗𝟙)▷⊸`, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the `⊸◁` bends the `a` strand down and the `▷⊸` brings it back up, while the `b` strand rides through untouched. Nothing else is spent below.

§4. ∩ is a commutative idempotent monoid on every hom-set

(4a)

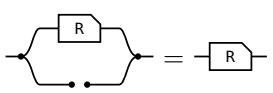
The **meet** of `R,S : a→b`, the paper's **convolution**, is `R∩S:=◁(R⊗S)▷` — copy the input, run `R` and `S` on the two copies, merge the results — so what comes out is what both of them do.



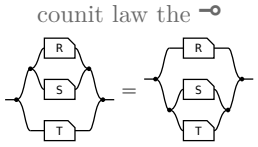
transcribed: a definition has no statement to export

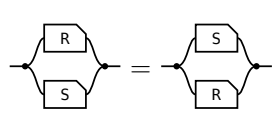
On every hom-set it is associative, commutative and idempotent , , with unit the maximal arrow `⊤=⊸∘⊸` (the paper's Lemma 4.11).

(4b)

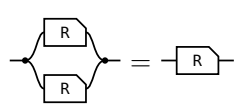


unit: one half of `⊤` per end — the merge's unit law absorbs the `⊸`, the copy's counit law the `⊸`





commutative: `σ` crosses `R⊗S` by naturality and is absorbed by cocommutativity and commutativity



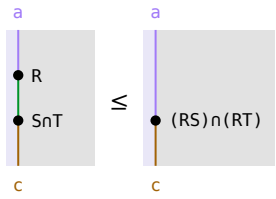
associative: coassociativity and associativity; `⊗` re-brackets for nothing, being strict here

idempotent: the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So `≤` is the order this monoid induces. `R∩S≤R` comes from the unit, and idempotency turns anything under both `S` and `T` into something under `S∩T`, since `R=R∩R≤S∩T`.

And one law relating ∩ to composition, which is **not** an equation:

(4c)

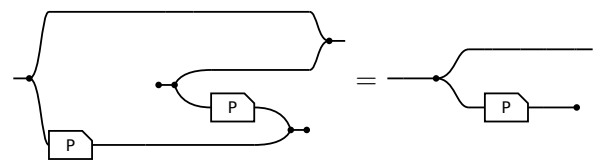
semi-distributivity, and what supplies it	**picture**
`R (S∩T)⊑RS∩RT` — the lax copy law. Equality exactly when `R` is single valued: the Maps section's `F(R∩S)=FR∩FS`.	

§5. Domain and range

(5a)

The **domain** `dom(R) ≜ 𝟙 ∩ (R R°)` and the **range** `ran(R) ≜ dom(R°)` .

(5b)



▷ lands the return leg back on the value ◁ handed out,

so R°▷ cuts to ⊸ — Frobenius

Running `R` and throwing the result away leaves only the fact that `R` could fire, and `ran(R)` is the same picture with the box mirrored. In `Rel` both steps are `{(a,a) : ∃b. a R b}`.

(5c)

1	`dom(R) ≜ 𝟙 ∩ (R R°)`
2	`Coreflexive(X) ⇒ dom(R) ⊑ X ⇔ R ⊑ X R`
3	`dom(R S) ⊑ dom(R)`
4	`dom(R ∩ S) = 𝟙 ∩ (S R°)`
5	`R` entire `Entire(R) ≜ dom(R) = 𝟙`
6	`R` simple `Simple(R) ≜ R° R ⊑ 𝟙`
7	`R` a map `Map(R) ≜ Entire(R) ∧ Simple(R)`

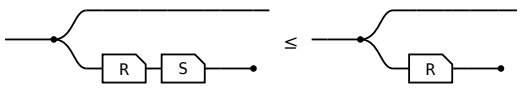
8	`Entire(R); Entire(S) ⇒ Entire(R S)`
	`Simple(f); Simple(g) ⇒ Simple(f g)`
	`Map(f); Map(g) ⇒ Map(f g)`

9	`Entire(R S) ⇒ Entire(R)`

§5.1. Sliding the discard

`dom(RS)⊑dom(R)`, and a single glyph for `dom` would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

(5.1a)



S⊸⊑⊸, the lax axiom for ⊸ in the first section — the discard slides back past S

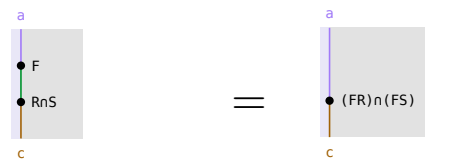
Equality is `S` entire, which is the same picture read as `dom(R)=𝟙⇔R` entire.

§6. Maps

(6a)

1	**surjective** $\mathbb{1} \sqsubseteq R^\circ R$, that is, R° entire.	**single valued** $R^\circ R \sqsubseteq \mathbb{1}$
2	**entire** $\mathbb{1} \sqsubseteq R R^\circ$. With **single valued**, a **map**.	**injective** $R R^\circ \sqsubseteq \mathbb{1}$

(6b)

law	picture
`Simple(F) ⇒ F (R ∩ S) = (F R) ∩ (F S)` F single valued	 one person who admires both = a at A, b at B

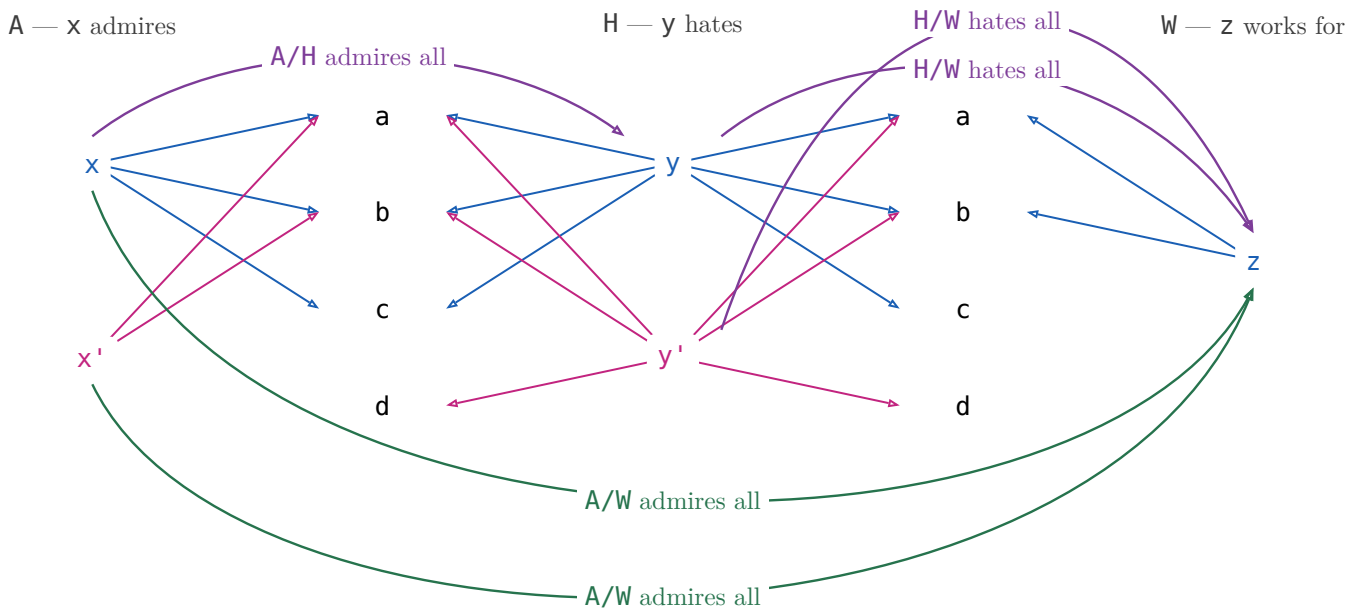
§8. / is all of

(8a)

```
x (R/S) y⇔∀p. y S p→x R p
x (S\R) y⇔∀p. p S x→p R y
/compares images: S(y)⊆R(x).   \ compares preimages: S°(x)⊆R°(y).
example: A admires,H hates,W works for
x (A/H) y — x admires everyone y hates.
x (H\A) y — everyone who hates x admires y.
```

§8.1. (R/S)(S/W)⊑R/W

(8.1a)

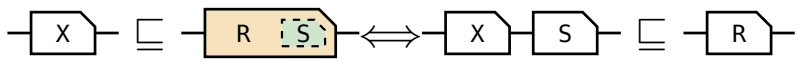
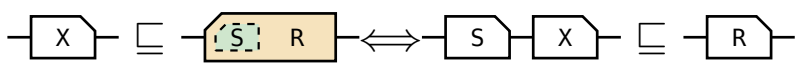
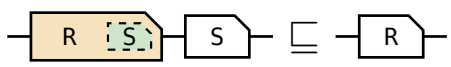
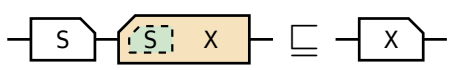
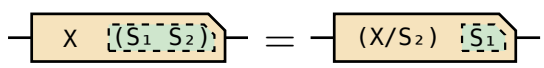
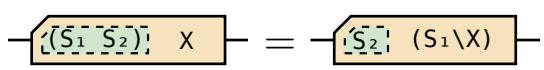
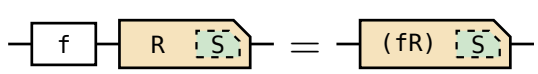
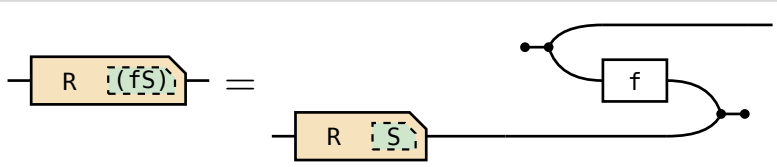
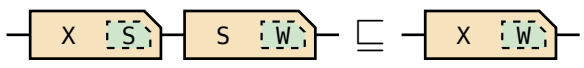
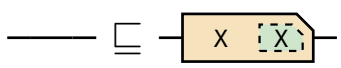
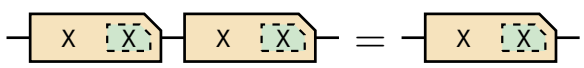
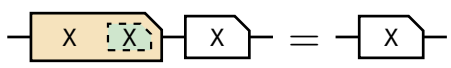
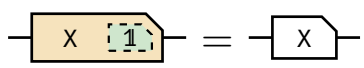
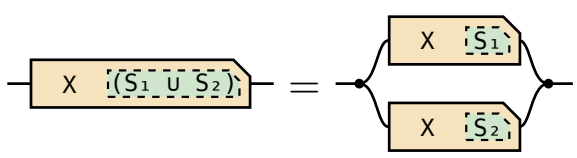
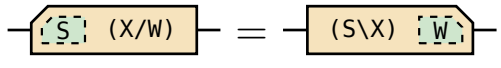


(8.1b)

x (A/H) y — x admires everyone y hates
y (H/W) z — y hates everyone z works for
x (A/W) z — x admires everyone z works for

(A/H)(H/W) is a path: x → y → z, and that is all of it. A/W also holds of x', who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of (R/S)(S/W)⊑R/W.

(8.1c)

1	X⊑R/S⇔XS⊑R X is any x-to-y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S, and R/S is the largest such.	
2	X⊑S\R⇔SX⊑R The mirror — divide on the left when x comes first.	
3	(R/S)S⊑R There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at S=∅: R/S is everyone, (R/S)S=∅.	
4	S (S\R)⊑R The mirror.	
5	**associate:** R/(S₁S₂)=(R/S₂)/S₁ **A friend's enemy** is two hops: divide by the far end first.	
6	(S₁S₂)\R=S₂\(S₁\R) The mirror.	
7	**maps:** f (R/S)=(fR)/S Rename x before or after dividing — the licence to write fR/S.	
8	R/(fS)=(R/S)f° Rename y: a map leaves a denominator as f° outside the box.	
9	(R/S)(S/W)⊑R/W Someone who admires all of a hate-set that already covers everyone z works for admires those people too.	
10	𝟙⊑R/R R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
11	(R/R)(R/R)=R/R R/R is the preorder **admires at least as much as**, and a preorder is idempotent. Freyd writes ⊑; with 𝟙⊑R/R above it is an equality.	
12	(R/R)R=R Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
13	R/𝟙=R Dividing by 𝟙: p's set is just {p}, so admiring all of it is admiring p.	
14	R/(S₁ ∪ S₂)=R/S₁∩R/S₂ Admiring a combined hate-set is admiring each set in full.	
15	S\(R/W)=(S\R)/W Which is why S\R/W needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: ∩, ∪, ∘ and composition are what the Frobenius generators build, and / is none of those — it is posited, with nothing to unfold.

§9. $\frac{R}{S}$

(9a)

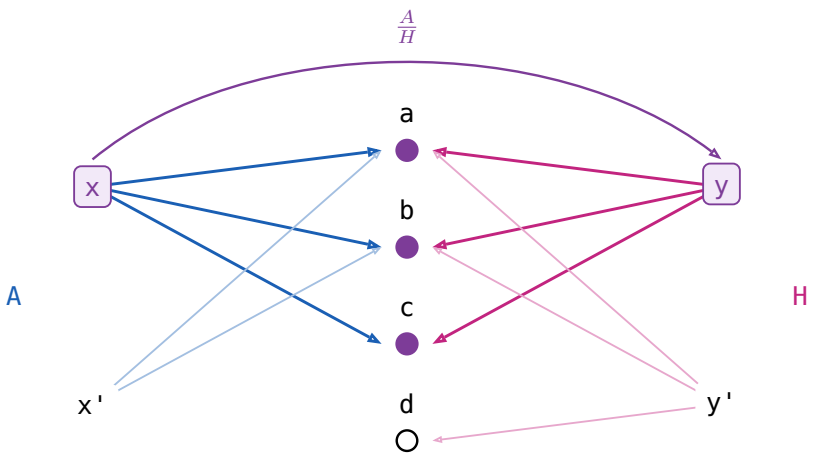
`R /ₛ S ≜ (R / S) ∩ (S / R)°` what `R` sends `x` to and nothing else is what `S` sends `y` to. In `Rel` `x` and `y` has the same image: `∀p. (x R p⇔y S p)`

(9b)

`x (A/H) y` — `x` admires everyone `y` hates
`x ((H/A)°) y` — `x` admires only people `y` hates
`x ((A/H)∩(H/A)°) y` — `x` admires only and all whom `y` hates

`x` admires exactly whom `y` hates: `/` is **all of**, $\frac{R}{S}$ is **only all of**.

(9c)



(9d)

	law	the reading
1	`X ⊑ R /ₛ S ⇔ X S ⊑ R ∧ X° R ⊑ S` `X` sits inside the matching exactly when it carries `S` back inside `R` and, reversed, carries `R` back inside `S`.	`X` may pair `x` with `y` only when `x` admires exactly whom `y` hates. Both halves must typecheck, so the operation is **partial**.
2	`(X /ₛ S)° = S /ₛ X` reversing the matching swaps the two relations.	Matching is symmetric.
3	`(X /ₛ S) (S /ₛ W) ⊑ X /ₛ W` a match from `R` to `S` followed by one from `S` to `W` is a match from `R` to `W`.	And transitive.
4	`(X /ₛ S) S ⊑ X` following the matching by `S` lands inside `R`.	$\left(\exists y.\, x\left(\frac{R}{S}\right)y \wedge ySp\right) \to xRp$ `x only admires whom y hates` $\frac{R}{S}S = \mathrm{Dom}\left(\frac{R}{S}\right)R$

5	`(X /ₛ X) X = X` matching `R` against itself and then following `R` gives `R` back.	$\left(\exists y.\, x\left(\frac{R}{R}\right)y \wedge yRp\right) \iff xRp$
		`x and y admire the same people`
		`y=x always qualifies (𝟙⊑R%R below)`

6	`𝟙 ⊑ X /ₛ X` every `x` matches itself.	$x\left(\frac{R}{R}\right)y$ if `x` and `y` admires the same peoples.
7	`(X /ₛ X)² = X /ₛ X` matching twice matches no more pairs than matching once.	So the relation **admires the same people** is an equivalence relation.
8	`Symmetric(X) ⇒ X ⊑ R /ₛ R ⇔ X R ⊑ R` a symmetric `X` sits inside the matching exactly when following it by `R` adds nothing to `R`.	The largest symmetric arrow that leaves `R` alone.
9	`simplePart(R) ≜ R /ₛ 𝟙` matching `R` against the identity keeps the `x` whose image is a single point.	The people who admire exactly one person and nobody else. It equals `R` only when `R` is simple, unlike `R/𝟙=R`.
10	`dom(X /ₛ S) = 𝟙 ∩ ((X / S) (S / X))` the `x` that match something are those the two one-sided divisions take round the loop back to `x`.	Its domain is the **domain of simplicity** of `R`.

§10. Freyd's Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows, ∋ **epsiloff**, subject to

1	$\ni_R \square = R\square, \quad \ni_R = \ni_R \square$	
2	$\mathbb{1} \sqsubseteq (R/\ni_R)(\ni_R/R)$	$\ni_R$ is **thick**
3	$\frac{\ni_R}{\ni_R} = \mathbb{1}$	$\ni_R$ is **straight**

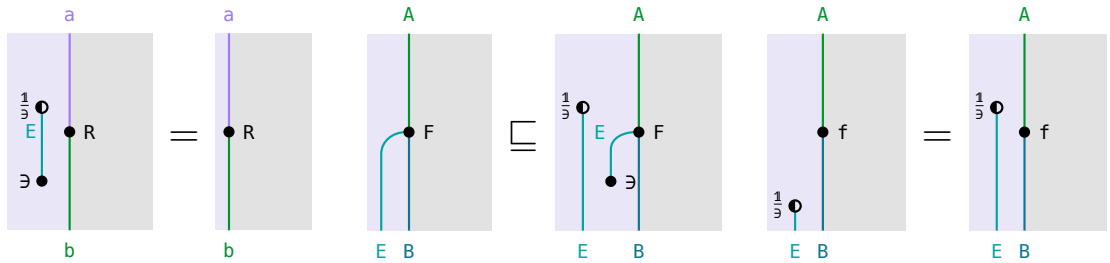
R□ is R's target, an identity arrow. For `R : A→B` write `∋ : PB→B`, dropping the subscript.

`∈≜∋° : A→PA`

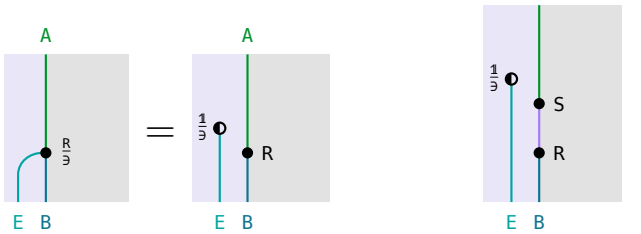
(10b)

	law	the reading
1	$\ni_R \square = R\square, \quad \ni_R = \ni_R \square$	One ∋ per object, not per arrow.
2	∋ is **thick**	**Comprehension**: every x has a set of exactly the people x admires.
3	$\frac{R}{\ni}$ `: A→PB`, for `R : A→B`	convert a relation to a function. `a` $\frac{R}{\ni}$ `={b|a R b}`
4	`Map(`$\frac{R}{\ni}$`)`	
5	$\frac{R}{\ni}$ `∋ = R`	reading the chosen set back through ∋ returns the relation.
6	`Simple(F) ⇒ F ⊑` $\frac{F\ \ni}{\ni}$	A partial choice of sets is inside the total one.
7	the **singleton map** is monic: $\frac{\mathbb{1}}{\ni}$ `(`$\frac{\mathbb{1}}{\ni}$`)° ⊑` $\mathbb{1}$ two points with the same one-person set are the same point.	The one-person set.
8	$\frac{\ni}{\ni}$ `= 𝟙`	Make the set of a set, then read it back one level down.
9	**fusion:** `Map(f) ⇒` $\frac{f\ R}{\ni}$ `=` $\frac{f}{\ni}$	Naturality of the unit, `Map(f) ⇒ f` $\frac{\mathbb{1}}{\ni}$ `=` $\frac{\mathbb{1}}{\ni}$ `E(f)` . renaming a point and then taking its one-person set is taking the set and renaming inside it.
10	`Map(f); 𝟙□ = ∋□ ⇒` $\frac{f}{\ni}$ `=` $\frac{\mathbb{1}}{\ni}$	Rename first or take singletons first — the fusion row above at R=𝟙.
11	`R□ = ∋□ ⇒ R /ₛ S =` $\frac{R}{\ni}$ `(`$\frac{S}{\ni}$`)°`	x and y match when R sends x and S sends y to the same set.
12	`E(R) ≜` $\frac{\ni\ R}{\ni}$	`E(R): PA→PB`, image of a set of A
13	$\frac{R}{\ni}$ `=` $\frac{\mathbb{1}}{\ni}$ `E(R)`	$\frac{\mathbb{1}}{\ni}$`: x↦{x}`
14	$\frac{S}{\ni}$ `E(R) =` $\frac{S\ R}{\ni}$	absorption — the monad's composition law, $\frac{S}{\ni} \diamond \frac{R}{\ni} = \frac{SR}{\ni}$, §10.3
15	`E(R) = E(`$\frac{R}{\ni}$`) union`	the image of a set under R is the union of the R-sets of its points.
16	`⊇ ≜ ∋ / ∋`	`xs⊇ys⇔∀a. ys∋a→xs∋a`
17	`⊆ = ⊇°`	`xs⊆ys⇔ys⊇xs`
18	`T°` $\frac{T}{\ni}$ `⊑ ∈`	every T-value of a point is a member of its T-set: `Λ(T)` is a map and `Λ(T)∋=T`.

(10c)



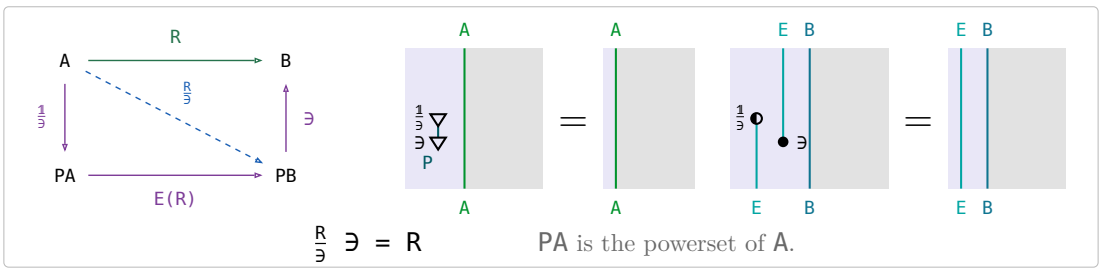
$\frac{R}{\ni}\ \ni = R$ $\quad$ Simple(F) ⇒ $F \sqsubseteq \frac{1}{\ni}\ E(F\ \ni)$ $\quad$ Map(f) ⇒ $f\ \frac{1}{\ni} = \frac{1}{\ni}\ E(f)$



$\frac{R}{\ni} = \frac{1}{\ni}\ E(R)$ $\quad$ $\frac{S}{\ni}\ E(R) = \frac{S\ R}{\ni}$

§10.1. i⊣E Power Allegory defined as adjunction

(10.1a)

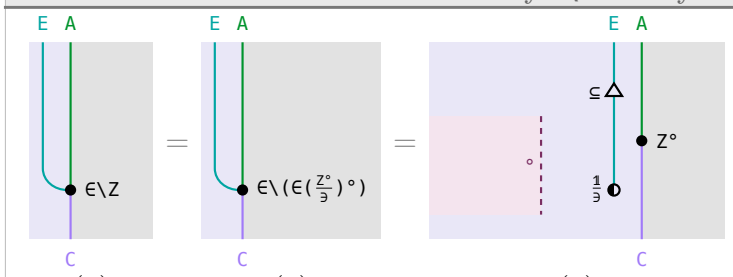


i is the inclusion Map(𝒜)→𝒜, doing nothing, so E is both the functor and the monad iE.

§10.2. ∈\ as a composite

$$\in \setminus Z = \subseteq (\tfrac{Z^\circ}{\ni})^\circ \qquad (10.2a)$$

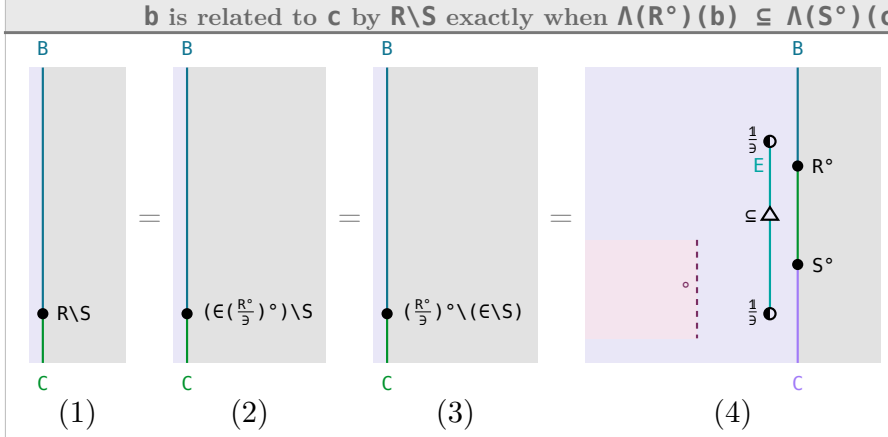
xs is related to c by ∈\Z exactly when xs ⊆ Λ(Z°)(c), the set of every a with a Z c



1	∈ \ Z	Z=∈Λ(Z°)° — R/∋ ∋ = R
2	= ∈ \ (∈ (Z°/∋)°)	X\(Yf°)=(X\Y)f°, f a map
3	= ⊆ (Z°/∋)°	

$$R \setminus S = \tfrac{R^\circ}{\ni} \subseteq (\tfrac{S^\circ}{\ni})^\circ \qquad (10.2b)$$

b is related to c by R\S exactly when Λ(R°)(b) ⊆ Λ(S°)(c), the R-preimage of b inside the S-preimage of c



1	R \ S	R=∈Λ(R°)° — R/∋ ∋ = R
2	= (∈ (R°/∋)°) \ S	(XY)\S=Y\(X\S)
3	= (R°/∋)° \ (∈ \ S)	f°\X=fX; ∈\S=⊆Λ(S°)° — ∈ \ Z = ⊆ (Z°/∋)°
4	= R°/∋ ⊆ (S°/∋)°	

§10.3. 𝒜≅Kleisli(E)

(10.3a)

E(R) ≜ $\frac{\ni R}{\ni}$, $\frac{\mathbb{1}}{\ni}$: A↛PA, union = E(∋) : PPA↛PA

f ⋄ g ≜ f E(g) union , for f : A↛PB and g : B↛PC

the monad is on Map(𝒜), not on the allegory: E is a relator on all relations, but $\frac{\mathbb{1}}{\ni}$, union and f E(g) union are maps, and the Kleisli construction happens where they live.

(10.3b)

$\frac{S}{\ni}$ ⋄ $\frac{R}{\ni}$ = (definition of ⋄, union=E(∋)) $\frac{S}{\ni}$ E($\frac{R}{\ni}$)E(∋) = (E a functor, E(X)E(Y)=E(XY)) $\frac{S}{\ni}$ E($\frac{R}{\ni}$∋)

= ($\frac{R}{\ni}$ ∋ = R , $\frac{R}{\ni}$ ∋=R) $\frac{S}{\ni}$ E(R) = ($\frac{S}{\ni}$ E(R) = $\frac{S}{\ni}\frac{R}{\ni}$, absorption) $\frac{SR}{\ni}$

the first three steps only unfold ⋄ — E is a functor, so E($\frac{R}{\ni}$)union=E($\frac{R}{\ni}$∋)=E(R) and the union is gone.
Absorption, $\frac{S}{\ni}$ E(R) = $\frac{S}{\ni}\frac{R}{\ni}$, is the whole law, and it is the functoriality of $\frac{\cdot}{\ni}$.

$\mathbb{1}$ goes to $\frac{\mathbb{1}}{\ni}$, the Kleisli identity, by definition — so $\frac{\cdot}{\ni}$ is an isomorphism of categories 𝒜≅Kleisli(E), and §10.1's i⊣E is the Kleisli adjunction. i is the identity on objects, so every object of the allegory is free.